Java Full Stack with Core-Java-Refresher

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Course type	**Hybrid**
Total Contact Hours	***132 Hours***

Recommended Semester	**8 Semester**

Course learning objectives	
The objective of the course is,	
1	Introduce core Java programming fundamentals, JVM architecture, object-oriented programming concepts, and modern Java (Java 8–17) features for building robust backend applications.
2	Develop enterprise application development skills using Spring Core, Spring Boot, Spring MVC, RESTful web services, ORM with Spring Data JPA, exception handling, validation, and security mechanisms.
3	Enable full-stack and cloud-ready application development by integrating frontend (ReactJS) with backend services, applying database management, unit testing, DevOps practices, microservices architecture, deployment strategies, and code quality tools.

Pre-requisites for the course	Basic knowledge of Computer and Operating Systems.

Day 1: Java Fundamentals + GitHub Basics (6 Hours)
• History & features • JVM, JRE, JDK • JVM architecture • Data types • Variables • Casting • Operators • Control flow (if, switch, loops, break, continue) • What is Git vs GitHub

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- Install Git
- Git operations (git init, git add, git commit, git status, git log)
- Create GitHub repo
- Push project to GitHub

Day 2: OOP & Class Design (6 Hours)

- Abstraction
- Encapsulation
- Inheritance
- Polymorphism
- Classes & constructors
- this & super
- Method overloading & overriding
- Interfaces
- Abstract classes
- SOLID principles (intro)

Day 3: Modifiers, Exception & File I/O Handling (6 Hours)

- Packages
- Access specifiers
- Modifiers (static, final, etc.)
- Exception handling
- Custom exceptions
- try-with-resources
- File I/O
- Streams

Day 4: Collections and Java 8 (6 Hours)

- Generics
- List, Set, Map
- Stack using Deque
- Comparable vs Comparator
- Functional programming

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- Functional interfaces
- Lambda expressions
- Streams API
- Optional
- Date & Time API

Day 5: Multithreading & Memory (6 Hours)

- Heap vs stack
- Garbage collection
- Thread lifecycle
- Synchronization
- Thread issues & deadlocks
- Executor framework

Day 6: JavaScript Foundations (6 Hours)

- JS syntax
- Variables (var, let, const)
- Operators
- Writing JS programs
- Console execution
- ES6 (arrow functions,Template literals, destructuring)
- Spread & rest operators
- Modules (import/export)
- Promises

Day 7: DOM & Asynchronous JavaScript (6 Hours)

- DOM structure
- Selecting elements
- Modifying HTML & CSS via JS
- Event handling
- Real-time DOM examples
- Fetch API
- Async / await

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- Handling API responses
- Error handling
- Intro to JS frameworks
- Why React?

Day 8: React Basics (6 Hours)

- React introduction
- JSX syntax
- Project structure
- Functional components
- Class components
- Props
- State (basic)
- Creating & updating state
- Class component state
- Lifecycle methods
- Hooks (UseState, useEffect)
- State Management (Context API, Redux)
- Component mounting & unmounting

Day 9: React Advanced & Mini Frontend (6 Hours)

- Controlled components
- Uncontrolled components
- Form validation & error messages
- React Router
- Route params
- State flow
- Error handling UI
- Small frontend mini project

Day 10: Spring Core (6 Hours)

- Spring framework overview
- IOC & DI

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- BeanFactory vs ApplicationContext
- XML-based configuration
- Bean scopes

Day 11: Annotations & Java Config (6 Hours)

- @Component
- @Service
- @Repository
- @Autowired
- @Qualifier
- @Value
- @Configuration
- @Bean
- AnnotationConfigApplicationContext

Day 12: Spring MVC & Spring Boot (6 Hours)

- DispatcherServlet
- Controllers
- Spring Boot overview
- Project creation
- Auto-configuration

Day 13: REST APIs (6 Hours)

- REST principles
- SOAP vs REST
- GET, POST, PUT, DELETE
- ResponseEntity
- Swagger documentation
- Global exception handling

Day 14: Validation & Security (6 Hours)

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- DTO validation
- Custom validators
- File upload & download
- Spring Security
- JWT authentication

Day 15: Microservices & Reactive (6 Hours)

- Monolith vs microservices
- Service registry & discovery
- API gateway
- Reactive programming
- WebFlux
- R2DBC

Day 16: JPA & Database (6 Hours)

- ORM concepts
- JPA annotations
- Spring Data JPA
- CRUD operations
- H2 database
- Relational DB concepts
- MySQL & engines
- Joins
- Indexes
- Transactions
- Spring Boot DB integration

Day 17: Full Stack Integration (6 Hours)

- React ↔ Spring Boot connection
- REST API consumption
- State flow
- Error handling

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- WebSockets

Day 18: Final Mini Project (6 Hours)

- Project implementation
- API testing
- Bug fixing
- Deployment
- Review & demo

Day 19: Advanced Git, Build Tools & GenAI (GitHub Copilot) (6 Hours)

- Git SSH login
- Merge
- Rebase
- Git reset & rollback strategies
- Advanced CI/CD integrations
- GitHub Actions & GitLab CI/CD overview
- Staging & branching
- Maven
- Gradle
- Copilot overview
- Code generation
- Debugging with Copilot
- Refactoring
- Best practices

Day 20: Docker, CI/CD & Cloud (6 Hours)

- Docker & Containerization
- DockerFile
- CI/CD overview
- Kubernetes
- Jenkins
- GitHub integration

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- Cloud services
- PaaS concepts
- Deployment models

Day 21: Testing, Code Quality & Security Tools (6 Hours)

- JUnit
- Mockito
- Unit testing
- Code coverage
- SonarQube
- Trivy
- OWASP Dependency Check
- Dockle

Day 22: Deployment (6 Hours)

- GitHub integration
- Frontend deployment (Netlify/Vercel)
- Backend deployment
- Environment configs

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Books	
	Reference Books:
1	**Effective Java** – Joshua Bloch
2	**Spring in Action** – Craig Walls
3	**Pro Git** – Scott Chacon & Ben Straub

Software Requirements :		
S.No	Software Taught	Versions available (Student/Paid/Free)
1	Java JDK (8 – 17)	Free
2	Eclipse / IntelliJ IDEA	Free
3	Spring Framework	Free
4	MySQL Database	Free
5	Git & GitHub	Free

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S.No	Software Required	Versions available (Student/Paid/Free)
1	Java JDK	Free
2	MySQL Server	Free
3	GitHub Account	Free
4	Docker Engine	Free

Projects	
S.No	Problem Statements
1	RESTful API Development using Spring Boot
2	CRUD Application with Spring Data JPA
3	JWT-based Authentication System
4	Exception Handling Framework
5	File Upload & Download Application
6	Microservices Communication using Spring Boot
7	API Documentation using Swagger
8	Database Integration with MySQL
9	Unit Testing using JUnit & Mockito
10	Dockerized Spring Boot Application
11	CI/CD Pipeline using Jenkins
12	SonarQube Code Quality Analysis

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13	Full Stack Integration (React + Spring Boot)
14	Secure REST API with Spring Security
15	End-to-End Full Stack Mini Project

Course Outcome (COs) At the end of the course, the student will be able to:		
Learning Levels: Re - Remember; Un - Understand; Ap - Apply; An - Analysis; Ev - Evaluate; Cr - Create		Learning Level
1	Understand Java programming fundamentals, JVM architecture, and object-oriented concepts	Un
2	Apply core Java concepts including collections, exception handling, multithreading, and file handling	Ap
3	Analyze application behavior using Java 8 features, streams, concurrency, and debugging tools	An
4	Develop enterprise applications using Spring Core, Spring Boot, RESTful services, and JPA	Cr
5	Implement secure, scalable, and testable full-stack applications with CI/CD and DevOps practices	Ap

Recommendations

- Practice Core Java concepts including OOPs, collections, and multithreading
- Build REST APIs using Spring Boot and test with Postman
- Apply Java 8 features such as streams, lambda expressions, and functional interfaces
- Implement proper exception handling and validation mechanisms
- Practice database integration using Spring Data JPA
- Secure applications using Spring Security and JWT

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- Write unit tests using JUnit and Mockito
- Perform code quality analysis using SonarQube
- Containerize applications using Docker
- Implement CI/CD pipelines using Jenkins
- Integrate frontend (ReactJS) with backend services
- Develop production-ready applications following best practices

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